Differential Effect of Electrical Stimuli on a 3D Bioprinted Model of Inflamed Skin

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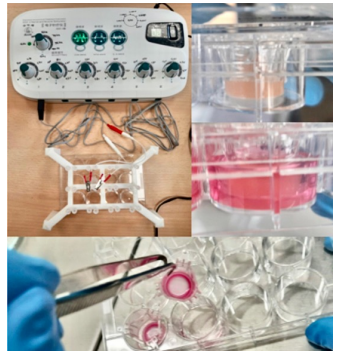
Introduction



3D construct containing fibroblasts and keratinocites in a collagen matrix S. Villata et. al, Biomedical Science and Engineering 2021; volume 4(s1):200

Electrostimulation is used in a variety of clinical approaches, However, magnitude and duration of the stimuli are not systematically tested, limiting standardization and mechanisms' elucidation. Consequently, therapeutic effectiveness remains unclear.

We designed **3D bioprinted constructs and explored the effect of the electrical stimulation in inflamed-like and physiological samples**, in the absence of nervous system, hypothesizing that functions involved in the *greater inflammatory pathway* (Maturo et al. 2020), i.e *wound healing (WH)*, are at stake.

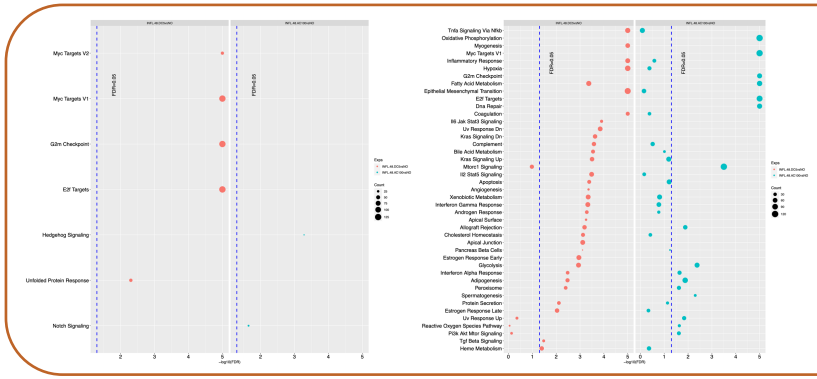
Methods

- 3D bioprinted construct of fibroblasts and keratinocytes (Villata et al. 2021);
- Inflammatory state mimicked with TNF-α sample perfusion;
- Sampling before/after stimuli (48h) in direct and alternate current;
- RNA-Seq data; DE analysis with DESeq2, exploratory enrichment with GSEA.

Results

Preliminary results show that there **exists a differential effect of the 5V direct current** (DC) stimulation versus no stimulus. In this experiment, the magnitude of the effect appears to be **more important with DC than AC at 100Hz**.

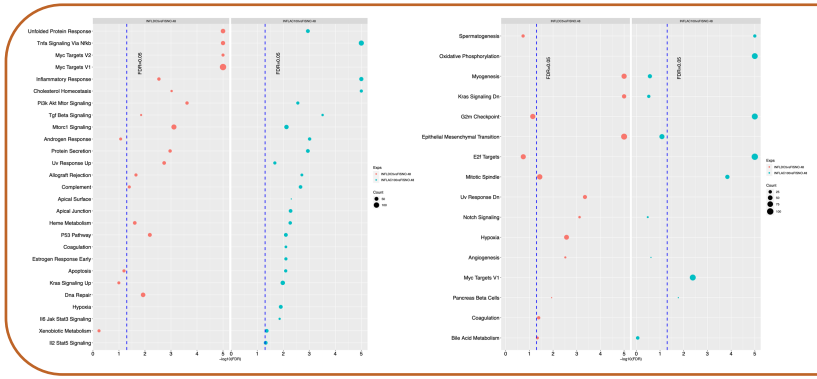
Based on our interest in WH, we focused our attention on the following GSEA *Hallmark* gene set, from which we retained *development*, *proliferation*, *immunity*, *signaling* and *pathway* categories



How do different stimuli affect inflamed samples?

DC: ↗ *proliferative* function, ↘ *inflammation* associated functions.

AC: less effective action.



Is the inflamed sample closer to physiology upon stimulus?

DC: ↗ proliferative pathways and inflammatory functions, ↘ development associated functions

AC: ↗ cellular components and inflammatory functions, ↘ proliferation associated functions

How do the effects of different stimuli differ?

↗ functions related to proliferation,

↘ immune response related to inflammation and developmental functions